📘 FLOW DOCUMENTATION

Product: ShipGen – Autonomous Logistics Operator

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1. 🎯 PURPOSE

Define **end-to-end system flow** from:

👉 Order → Trip → Execution → Delivery → Profit

With:

- AI decision points
- Automation triggers
- User interaction points

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2. 🧠 CORE FLOW PHILOSOPHY

> **"System acts first, user intervenes only if needed."**

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3. 🔄 END-TO-END FLOW (HIGH LEVEL)

1. Order Received
2. Order Processed (AI)
3. Trip Auto Created
4. Driver & Vehicle Assigned
5. Route Optimized
6. Trip Approved (or auto-approved)
7. Execution Started
8. Tracking + Prediction
9. Exception Handling
10. Delivery Completed
11. Profit Calculated

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4. 📩 STAGE 1: ORDER INGESTION FLOW

Trigger:

- Email / WhatsApp / API

Steps:

1. Raw message received
2. NLP Engine processes text
3. Extract:
    - Pickup location
    - Drop location
    - Load details
    - Date

AI Role:

- Natural Language Processing (NLP)

Output:

👉 Structured Order Object

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5. 🧠 STAGE 2: ORDER VALIDATION FLOW

Steps:

1. Validate extracted data
2. Check missing fields

Decision:

- If complete → proceed

- If incomplete →
  👉 send clarification request to user

AI Role:

- Context understanding

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6. 🚚 STAGE 3: AUTO TRIP CREATION FLOW

Steps:

1. Trigger Trip Engine
2. Fetch:
   - available drivers
   - available vehicles
3. Run matching algorithm

AI Decisions:

- Best driver selection
- Best vehicle selection
- Cost estimation

Output:

👉 Trip Object (READY STATE)

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7. 🛣️ STAGE 4: ROUTE OPTIMIZATION FLOW

Steps:

1. Call Maps API
2. Calculate routes

AI Decisions:

- Shortest route
- Traffic-aware route
- Alternate route

Output:

👉 Route Plan attached to Trip

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8. ✅ STAGE 5: APPROVAL FLOW

Options:

Mode 1: Manual Approval

- User reviews trip
- Clicks “Approve”

Mode 2: Auto Approval

- System auto-confirms based on rules

Output:

👉 Trip moves to ACTIVE state

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9. 📞 STAGE 6: EXECUTION FLOW

Steps:

1. Driver notified

2. Customer notified
3. Trip instructions shared

Communication:

- WhatsApp / SMS / App

Automation:

- Retry if no response
- Escalate if driver inactive

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10. 📡 STAGE 7: LIVE TRACKING FLOW

Steps:

1. GPS data received
2. Location updated
3. Event generated

Events:

- Trip started
- Reached pickup
- In transit
- Delivered

Output:

👉 Real-time tracking

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11. 🔮 STAGE 8: PREDICTION FLOW

Inputs:

- GPS data
- Route
- Historical data

AI Decisions:

- Delay probability
- ETA updates

Output:

- “On time”
- “Delay risk”

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12. 🚨 STAGE 9: EXCEPTION HANDLING FLOW

Triggers:

- Delay detected
- Driver stopped
- Route deviation

Steps:

1. Alert generated
2. AI suggests action:
    - reroute
    - reassign driver
3. User notified (if needed)

Output:

👉 Issue resolved automatically / with approval

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13. 📦 STAGE 10: DELIVERY FLOW

Steps:

1. Driver marks delivery complete
2. System verifies location

Output:

👉 Trip status = COMPLETED

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14. 💸 STAGE 11: PROFIT CALCULATION FLOW

Steps:

1. Calculate:
    - fuel cost
    - driver cost
    - tolls
2. Compare with revenue

Output:

👉 Profit report

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15. 🔁 STAGE 12: LEARNING LOOP (AI IMPROVEMENT)

Steps:

1. Store trip data
2. Analyze:
    - delays
    - driver performance

AI Role:

- Improve future decisions
- Optimize assignments

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16. 🔄 COMPLETE FLOW SUMMARY

Order →
AI extracts data →
Trip auto created →
Driver assigned →
Route optimized →
Trip approved →
Driver notified →
Tracking starts →
AI predicts issues →
System resolves →
Delivery completed →
Profit calculated →
System learns

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17. 👥 USER INTERACTION POINTS

User only interacts at:

- Approval stage
- Exception handling (optional)

Everything else:

👉 Fully automated

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18. ⚡ AUTOMATION TRIGGERS

- New Order → Auto Trip
- Trip Approved → Auto Execution
- Delay Detected → Auto Alert
- Delivery Complete → Auto Profit

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19. 💣 CRITICAL DIFFERENTIATOR FLOW

Traditional:

Order → User creates trip → assigns → tracks

ShipGen:

Order → **System does everything → User approves**

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20. 🚀 FINAL FLOW STATEMENT

> "From order intake to delivery completion, every step is system-driven, with minimal human intervention."